MIL-C-11053F(ME)
28 December 1983
SUPERSEDING
MIL-C-11053E(ME)
25 July 1977

MILITARY SPECIFICATION

COUPLING ASSEMBLIES, HOSE

(CHEMICAL, 3/4 AND 1-INCH)

This specification is approved for use by the Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types and two sizes of high pressure hose coupling assemblies for connecting chemical or fire hose.

1.2 Classification. Coupling assemblies shall be the following types and sizes, as specified (see 6.2).

Type II - Expansion ring.

Type III - Compression wedge.

Size - 3/4 - inch (inside diameter).

Size - 1 - inch (inside diameter).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4210

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-775	- Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles and Strainers.
MIL-R-11521	- Rings, Expansion, Hose, Brass.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-H-52992	- Hose and Hose Assemblies, Chemical (Industrial and Fire Fighting).
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28/10	- Screw-Thread Standards for Federal Services, Section 10 American National Hose Couplings and Fire-Hose Coupling Threads.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

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(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 16 - Free-Cutting Brass Rod, Bar, and Shapes for Use In Screw Machines.
ASTM B 283 - Copper and Copper-Alloy Die Forgings (Hot-Pressed).
ASTM B 584 - Copper Alloy Sand Castings for General Applications.

(Application for copies should be addressed to the American Society For Mechanical Engineers, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 194 - Standards for Screw Threads and Gaskets for Fire Hose Connections.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Square, Quincy, MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The coupling assemblies shall be bronze castings or brass forgings, with brass fittings, for connecting high-pressure chemical and fire hoses at pressures up to 600 pounds per square inch (psi). The coupling assemblies shall accommodate 3/4-inch or 1-inch hose, as applicable, conforming to MIL-H-52992. The coupling assemblies shall have round bores of minimum diameter for full flow capacity for the applicable hose.

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3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (6.4).

3.3.1 Material deterioration and control. The coupling assemblies shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Bronze castings. Bronze castings shall conform to ASTM B 584, Alloy UNS C92200.

3.3.3 Brass. The brass shall conform to ASTM B 16, Alloy UNS C36000 or ASTM B 283, Alloy UNS C37700, as applicable.

3.4 Gaskets. Gaskets shall be provided with the couplings and shall conform to NFPA No. 194.

3.5 Threads. Threads shall be American National Fire-Hose Coupling Threads conforming to FED-STD-H28/10.

3.6 Lugs. Coupling assemblies shall be provided with wrenching lugs, for 3/4-inch and rocker lugs for the 1-inch size.

3.7 Finish. Coupling assemblies shall have a smooth finish in accordance with the contractor's standard practice.

3.8 Marking. Coupling assemblies shall be marked for their size and manufacture by stamping or embossing in accordance with the contractor's standard practice.

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3.9 Types of hose assemblies.

3.9.1 Type II, expansion ring. Type II coupling assemblies shall be provided with internally corrugated sockets and mated expansion rings, to be assembled as shown in figure 1 to produce a leak-free connection.

3.9.1.1 Expansion rings. Expansion rings shall conform to MIL-R-11521 for the applicable size.

3.9.2 Type III, compression wedge. Type III coupling assemblies shall be provided with corrugated sockets and threaded expanded wedges to be assembled as shown in figure 2 to produce a leak-free connection. Internal cylinder key-type wrenches shall be furnished in the quantity as specified (see 6.2).

3.10 Workmanship. Coupling assemblies shall be of uniform dimension for an applicable type, clean and free of dirt, chips, and scale. External surfaces shall be free from burrs, sharp edges, and corners except where required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article coupling assemblies shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection of the applicable hose assembly.

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4.3.2 Test. The first article coupling assemblies shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection of the applicable coupling assembly.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105, inspection level S-3.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 10.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The sample coupling assemblies shall be examined as specified herein, for the following defects:

Major

101. Material not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatment.
105. Bronze castings not as specified.
106. Brass not as specified.
107. Gaskets not as specified.
108. Threads not as specified.
109. Lugs not as specified.
110. Finish not as specified.
111. Type II, expansion ring not as specified.
112. Expansion rings not as specified.
113. Type III, compression wedge not as specified.

Minor

201. Marking not as specified.
202. Workmanship not as specified.

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4.5.2 Pressure test. Connect the coupling assembly to a section of applicable hose conforming to MIL-H-52992 as recommended by the contractor. Subject the coupling assembly to an internal pressure of 600 psi, then submerge in water for a period of not less than 3 minutes. Evidence of leakage at either end of the hose and coupling assembly, or from the mating parts of the assembly, or any creeping, loosening or separating of the hose and assembly shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing and marking for level A or B shall be examined to determine compliance with MIL-P-775. The preservation, packing and marking for commercial shall be examined in accordance with MIL-STD-1188.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Coupling assemblies shall be preserved in accordance with the level A preservation and packaging requirements of MIL-P-775.

5.1.2 Commercial. Coupling assemblies shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A or B or commercial as specified (see 6.2).

5.2.1 Level A. Coupling assemblies preserved as specified in 5.1, shall be packed in accordance with the level A packing requirements of MIL-P-775.

5.2.2 Level B. Coupling assemblies, preserved as specified in 5.1, shall be packed in accordance with the level B packing requirements of MIL-P-775.

5.2.3 Commercial. Coupling assemblies, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packing shall be in accordance with MIL-STD-1188.

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6. NOTES

6.1 Intended use. The coupling assemblies are intended for use with a hose used in handling water and chemicals for fire-fighting equipment.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size of coupling assembly required (see 1.2).

c. When a first article is required for inspection and approval (see 3.2).

d. Number of key-type wrenches required (see 3.9.2).

e. Degree of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction model coupling assembly. The first article should consist of one or more coupling assemblies. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, tests, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

6.5 Classification change. Type I and type IV coupling assemblies have been discontinued since they are no longer required.

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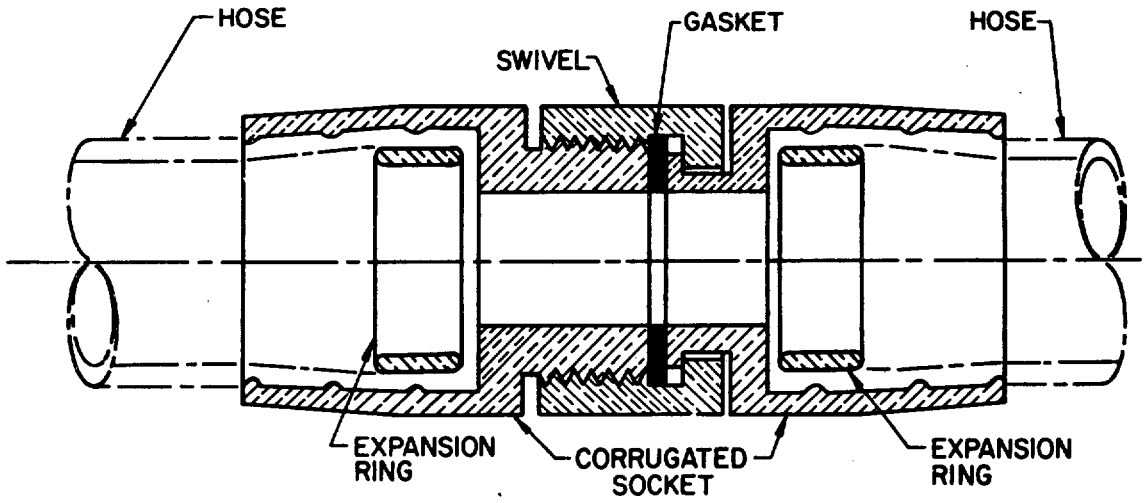


FIGURE 1. Type II, expansion ring.

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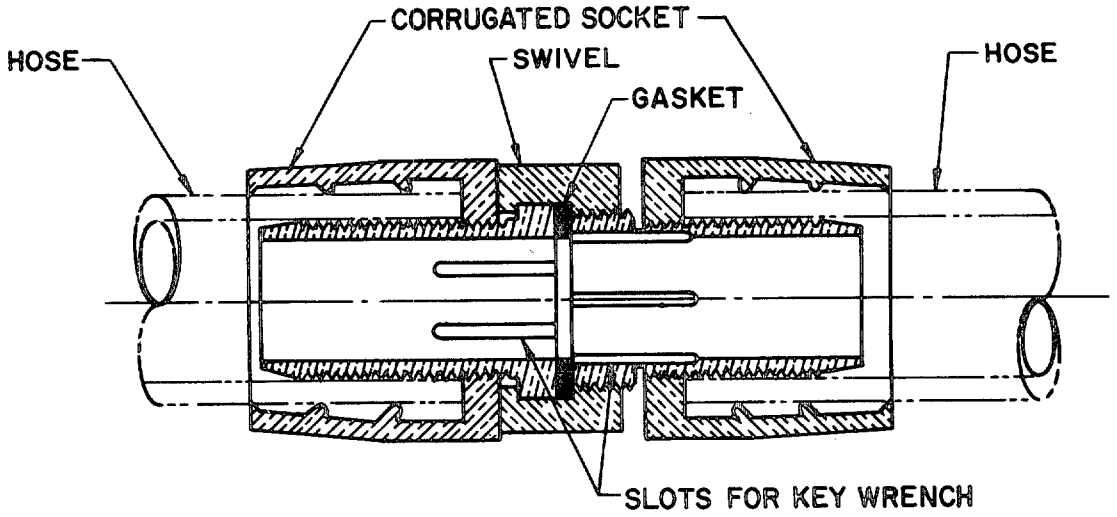


FIGURE 2. Type III, compression wedge.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-11053F(ME)	Coupling Assemblies, Hose (Chemical, 3/4 and 1 Inch)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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